

[This question paper contains 4 printed pages.]

Sr. No. of Question Paper : 779

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Your Roll No.....

Unique Paper Code : 234501

Name of the Paper : Theory of Computation (CSHT-511)

Name of the Course : B.Sc. (H) Computer Science

Semester : V

Duration : 3 Hours

Maximum Marks : 75

Instructions for Candidates

1. Write your Roll No. on the top immediately on receipt of this question paper.
2. All questions from Part A is compulsory and attempt any four questions from Part B.
3. Assume $\Sigma = \{a,b\}$ is the underlying alphabet unless mentioned otherwise. Parts of a question must be answered together.

PART A

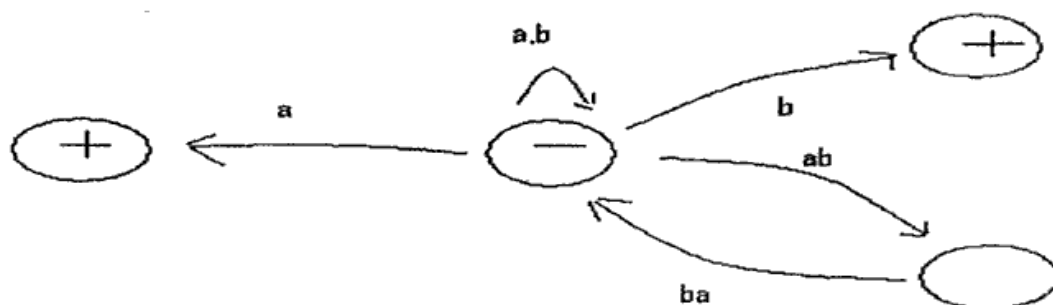
1. (a) Prove that for all sets S , $(S^+)^+ = S^+$. (2)
(b) Give regular expression for the language of all words that have at least two a's in them. (2)
(c) Consider the language PALINDROME over the alphabet $\{a b\}$. Prove that if x is in PALINDROME then so is x^n for any n . (3)
(d) Show that $(a^+b)^*$ and $(a+b)^*$ defines the same language over alphabet $\{a b\}$. (3)
(e) Build an FA that accepts only those words that have more than four letters. (3)

P.T.O.

- (f) Build FA for the regular expression $(a+b)b(a+b)^*$. (3)
- (g) Find a CFG for the language defined by regular expression $(baa + abb)^*$. (3)
- (h) Use the pumping lemma to show that the language $\{a^n b^n a^n \mid n = 1, 2, 3, \dots\}$ is non regular. (4)
- (i) Show that if L_1 and L_2 are regular languages, then so are $L_1 + L_2$, $L_1 L_2$ and L_1^* . (4)
- (j) Construct a PDA for the language $L = \{a^{2n} b^n \mid n = 0, 1, 2, 3, \dots\}$. (4)
- (k) Explain the Church Turing Thesis. (4)

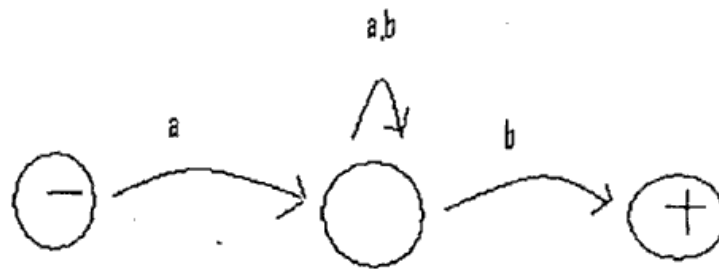
PART B

2. (a) Define Finite Automata. (2)
- (b) Build a regular expression for all words that have exactly two b's or exactly three b's not more. (3)
- (c) Build an FA that accepts only those words that begin or end with double letter. (5)
3. (a) Define Non Deterministic Finite Automaton. (2)
- (b) Convert the following Transition graph into regular expression. (4)



(c) Convert the following NFA into DFA :

(4)



4. (a) For the given languages $L_1 = (a+b)b(a+b)^*$ and $L_2 = b(a+b)^*$, find regular expression and finite automata that define $L_1 \cap L_2$. (5)
- (b) Use pumping lemma to show that language $\{a^{2n}b^n \mid n = 1, 2, 3, \dots\}$ is non regular. (5)
5. (a) Construct a CFG for the language $L = \{a^m b^n \mid n > m, m, n \geq 1\}$. (5)
- (b) Construct a PDA for the language $L = \{a^n b b^n \mid n = 1, 2, 3, \dots\}$. (5)
6. (a) State pumping lemma for context free languages. (2)
- (b) Show that the family of context free languages is not closed under intersection. <https://www.dudelhi.com> (4)
- (c) Show that the language $\{a^n b^n a^n b^n a^n \mid n = 1, 2, 3, \dots\}$ is non context free. (4)
7. (a) Define Turing Machine. (2)
- (b) Prove that If L is a recursive language, then its compliment L' is also recursive. (4)

P.T.O.

- (c) Design a Turing Machine that provides output as a compliment of the given number which is provided to the machine as input in binary form over the alphabet $\{01\}$. (4)

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